Friedrich-Alexander-Universität
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Practical Skills in Computational Biology

Fundamentals of Bioimage Processing - Unit 1

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2. NumPy Fundamentals

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From Continuous to Digital Images

- Real-world image (continuous signal):

$$I(x, y) \in \mathbb{R}$$

- Digital image:
  - Discrete spatial sampling:

$$I[x, y], \quad x, y \in \mathbb{Z}$$

  - Quantized intensity values with bit depth b:

$$I[x, y] \in \{0, \ldots, 2^b - 1\}$$

- Two key steps:
  - Sampling (space)
  - Quantization (intensity)

Sampling: Continuous to Discrete Domain

- Continuous signal:

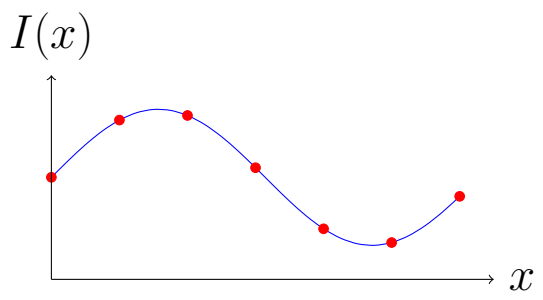
$$I(x) \in \mathbb{R}$$

- Sampling converts it to discrete points:

$$I[x], \quad x \in \mathbb{Z}$$

- Only values at specific positions are retained
- Controlled by the **sampling rate**

Sampling Example (1D Signal)



Blue: continuous signal Red: sampled points

Quantization: Continuous to Discrete Values

- Sampled signal still has continuous values:

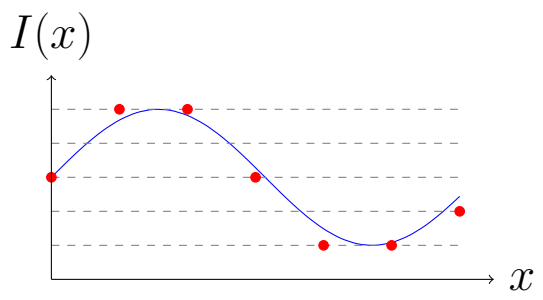
$$I[x] \in \mathbb{R}$$

- Quantization maps values to finite levels:

$$I[x] \in \{0, \ldots, 2^b - 1\}$$

- Depends on **bit depth** b
- Introduces **quantization error**

Quantization Example (1D Signal)



Blue: original signal Red: quantized values

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Efficient Numerical Computing with NumPy

- NumPy provides the `ndarray` class for multidimensional arrays.
- Key advantages:
  - Efficient memory representation
  - Fast operations (implemented in C/Fortran)
  - Vectorized computations
- Standard import:
  - `import numpy as np`

Properties of NumPy Arrays

- **shape**: dimensions of the array
- **dtype**: data type (e.g., uint8, float32)
- **ndim**: number of dimensions

Example:

Image shape: (X, Y) or (X, Y, C) or (T, Z, C, X, Y)

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Reading Images

- Images can be loaded using libraries such as `scikit-image`.
- Example function:
  - `skimage.io.imread()`
- Supports:
  - Local files
  - Relative paths
  - URLs

Displaying Images

- Common visualization tools:
  - `matplotlib.pyplot.imshow`
  - Specialized tools for microscopy images
- Important considerations:
  - Intensity scaling
  - Multi-channel handling
  - Colormaps (lookup tables)

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Indexing

- Access individual pixels or elements:

$$I[x, y]$$

- Higher dimensions:

$$I[t, z, c, x, y]$$

- Properties:
  - Returns a single value or lower-dimensional array
  - Zero-based indexing (NumPy)

Slicing

- Extract subarrays using index ranges:

$$I[a : b,\ c : d]$$

- Interpretation:
  - Rows: a to b
  - Columns: c to d
- Properties:
  - Returns a **view** (no data copy)
  - Supports step sizes: $I[a : b : s, c : d : t]$

Cropping

- Cropping = selecting a region of interest (ROI):

$$I_{\text{crop}} = I[a : b,\ c : d]$$

- Purpose:
  - Focus on relevant image regions
  - Reduce data size
- Typical use:
  - Object extraction
  - Preprocessing step

Masking

- Define a boolean mask:

$$M[x, y] = (I[x, y] > T)$$

- Apply mask:

$$I_{\text{masked}} = I[M]$$

- Applications:
  - Thresholding
  - Region selection

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Pixel-wise Arithmetic

- Images are numerical arrays:

$$I[x, y] \in \mathbb{R}$$

- Operations are applied element-wise:

$$I_{\text{new}}[x, y] = f(I[x, y])$$

- Examples:
  - Addition: $I + c$
  - Scaling: αI
  - Nonlinear: $\exp(I)$

Combining Images

- Combine images of same shape:

$$I_{\text{result}}[x, y] = I_1[x, y] + I_2[x, y]$$

- General form:

$$I_{\text{result}} = \alpha I_1 + \beta I_2$$

- Applications:
  - Image blending
  - Noise reduction (averaging)

Statistical Measures

- Number of pixels: N
- Mean intensity:

$$\mu = \frac{1}{N} \sum_{x,y} I[x, y]$$

- Variance:

$$\sigma^2 = \frac{1}{N} \sum_{x,y} (I[x, y] - \mu)^2$$

- Min / Max:

$$\min_{x,y} I[x, y], \quad \max_{x,y} I[x, y]$$

Histograms

- Distribution of intensity values
- Frequency of value k:

$$h[k] = \sum_{x,y} \mathbf{1}\{I[x,y] = k\}$$

- Normalized histogram:

$$p[k] = \frac{h[k]}{N}$$

- Interpretation:
  - Contrast
  - Dynamic range usage

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References

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